ELEC5305 Project Proposal

STFT-Based Speech Noise Reduction: Evaluation and Modification of Spectral Subtraction

1. Student Information

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2. Project Overview

Background noise can reduce the intelligibility and quality of recorded speech. This project will investigate Short-Time Fourier Transform (STFT)-based spectral subtraction using a small controlled experiment. A basic spectral subtraction system will first be implemented and evaluated. Its main shortcomings will then be identified, followed by a modified version using an over-subtraction factor and a spectral floor.

The research question is: Across white and non-stationary environmental noise at input SNRs of 0, 5, and 10 dB, how do basic and modified spectral subtraction compare in SNR improvement and STOI, and what trade-off between residual noise and speech distortion is introduced by the over-subtraction factor and spectral floor?

3. Background and Motivation

Speech is non-stationary, so its frequency content changes over time. The STFT is suitable for speech processing because it analyses short overlapping frames. Each frame is windowed, transformed to the frequency domain, modified, and reconstructed using the inverse STFT and overlap-add.

Spectral subtraction estimates a background-noise spectrum and subtracts it from the magnitude spectrum of noisy speech while retaining the noisy phase for reconstruction [1]. It is simple and interpretable, but inaccurate noise estimation or aggressive subtraction can leave isolated spectral peaks that create musical-noise artefacts, while excessive attenuation can distort speech. Over-subtraction and spectral flooring are established modifications intended to reduce these artefacts [2].

Modern speech enhancement increasingly uses data-driven systems and large-scale noisy speech datasets [4], [5]. However, a classical STFT-based method is appropriate here because the effect of each processing parameter can be examined directly. The project therefore follows a focused cycle: implement a baseline, evaluate it, modify it in response to observed shortcomings, and re-evaluate it under the same conditions.

4. Experimental Data

All audio will be processed at 16 kHz in mono. Twelve clean English utterances, approximately 3-8 seconds each, will be selected from the LibriSpeech test-clean corpus. Two speakers will be used, with six utterances from each. Four utterances, two per speaker, will form a development subset; the remaining eight will form an independent test subset.

Two noise conditions will be used: synthetic white Gaussian noise and one non-stationary environmental noise recording from the MUSAN noise corpus. Each utterance will be mixed with each noise type at 0, 5, and 10 dB input SNR. This gives 72 mixtures: 24 development mixtures and 48 final test mixtures. A 0.5-second noise-only segment will precede the speech in every mixture so that noise estimation is consistent.

5. Proposed Methodology

MATLAB will be the main implementation platform. STFT parameters will be fixed: a 512-sample Hann window (32 ms), a 256-sample hop size (50% overlap), and a 512-point FFT. Fixing these values keeps the project focused on spectral subtraction rather than broad parameter optimisation.

For the baseline, the average magnitude spectrum of the initial 0.5-second noise-only frames will be used as the noise estimate:

$$|S\_base(m,k)| = max(|Y(m,k)| - |N\_hat(k)|, 0)$$

where Y(m,k) is the noisy STFT and N_hat(k) is the estimated noise magnitude. The enhanced complex spectrum will retain the noisy phase, and the signal will be reconstructed using inverse STFT and overlap-add.

The modified method will use:

$$|S\_mod(m,k)| = max(|Y(m,k)| - alpha|N\_hat(k)|, beta|N\_hat(k)|)$$

where alpha controls over-subtraction and beta defines the spectral floor. Only alpha and beta will be tuned. They will be selected using the 24 development mixtures and then fixed before final evaluation on the 48 test mixtures. Baseline and modified systems will therefore be compared on identical test mixtures, avoiding parameter selection on the final test data.

6. Evaluation

Two formal metrics will be reported. First, SNR improvement will be calculated as output SNR minus input SNR. With the clean reference available:

$$SNR\_out = 10\ log10(\ sum\ s[n]\text{^}2\ /\ sum\ (s[n] - s\_hat[n])\text{^}2\ )$$

where s[n] is clean speech and s_hat[n] is the enhanced output. This reflects both residual noise and speech distortion relative to the clean reference.

Second, Short-Time Objective Intelligibility (STOI) will be used to assess speech intelligibility [3]. Results will be reported by noise type and input SNR, with an overall paired comparison between the baseline and modified methods. Representative waveforms and spectrograms will support the quantitative analysis. Informal listening will be used to describe residual noise, musical-noise artefacts, and audible speech distortion, but will not be treated as a formal subjective test.

7. Expected Outcomes

The project will produce a MATLAB baseline, a modified spectral subtraction implementation, held-out test results, SNR-improvement and STOI comparisons, representative waveform and spectrogram figures, and audio examples. The main outcome will be an evidence-based assessment of whether the modified method offers a useful trade-off between noise reduction and speech distortion. Code, results, and demonstrations will be documented on the GitHub project site, with a final report and video demonstration.

8. Timeline

Weeks 1-2: Define the project task and scope.

Weeks 3-5: Literature review, dataset selection, and preparation.

Week 6: Implement the STFT/iSTFT framework.

Week 7: Implement baseline spectral subtraction.

Week 8: Evaluate baseline shortcomings.

Week 9: Tune alpha and beta on the development subset.

Week 10: Finalise the modified method.

Week 11: Evaluate both methods on the fixed test subset.

Week 12: Analyse results and limitations.

Week 13: Complete the report, GitHub documentation, and video.

9. References

[1] S. F. Boll, "Suppression of Acoustic Noise in Speech Using Spectral Subtraction," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 27, no. 2, pp. 113-120, 1979.

[2] M. Berouti, R. Schwartz, and J. Makhoul, "Enhancement of Speech Corrupted by Acoustic Noise," Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 1979.

[3] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, "An Algorithm for Intelligibility Prediction of Time-Frequency Weighted Noisy Speech," IEEE Transactions on Audio, Speech, and Language Processing, vol. 19, no. 7, pp. 2125-2136, 2011.

[4] C. K. A. Reddy et al., "The INTERSPEECH 2020 Deep Noise Suppression Challenge: Datasets, Subjective Testing Framework, and Challenge Results," Proceedings of Interspeech, 2020.

[5] H. Schroter et al., "DeepFilterNet: A Low Complexity Speech Enhancement Framework for Full-Band Audio Based on Deep Filtering," Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2022.